Sudarshan Sudhakar

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Stealthy Metering Integrity Attack Detection in Smart EV Charging Infrastructure Using Attention-Based Bidirectional LSTM with Adaptive Isolation Forest–CUSUM

Sudarshan Sudhakar CS23B2007

***Abstract*—False Data Injection (FDI) attacks on Electric Vehicle (EV) charging infrastructure represent a critical class of cyberthreat that simultaneously compromises billing integrity, grid scheduling accuracy, and demand-response reliability. Existing detection schemes rely either on simplistic statistical thresholds that fail under low-amplitude perturbations, or on supervised classifiers that require labelled attack data unavailable in real deployments. This paper proposes a comprehensive extension of the NARX-based FDI detection framework to the large-scale, publicly available ACN-Data-Static dataset, comprising over 85 000 charging sessions from three California sites. Two complementary neural energy estimators are investigated: a *Nonlinear Auto-Regressive with eXogenous inputs* (NARX) network (171 parameters) and an *Attention-enhanced Bidirectional LSTM* (Attention-BiLSTM) (≈2.16 M parameters). Both estimators feed into a novel *adaptive two-stage Isolation Forest–Cumulative Sum (IF–CUSUM)* detector that augments the conventional one-dimensional Error-of-Estimation (EoE) signal with four engineered temporal features $[\varepsilon, \varepsilon^2, |\Delta\varepsilon|, z_{\text{loc}}]$ and calibrates the CUSUM threshold via a *distribution-shift cap* $h_{\text{cap}} = 3.5\, p_{99}(\text{EoE}_{\text{clean}})$. This cap prevents the detector from over-fitting to stronger attacks present in the validation slice, a systematic bias that silently suppressed recall by up to 29 percentage points in prior work. The NARX-based system achieves an F1-score of 97.9% with recall of 99.9% and precision of 96.0% on realistic low-amplitude attacks ($\phi \in [1, 2]$). The Attention-BiLSTM variant yields 97.0% F1 with 98.7% recall. A naive mean-predictor baseline records only 35.6% F1, confirming that estimator quality—not the downstream detector—is the primary performance driver. An ablation study quantifies the individual contributions of the 4-D feature set and the distribution-shift cap. All experiments use provably zero session overlap between training and evaluation sets.**

***Index Terms*—False data injection attack, EV charging network, NARX neural network, attention-bidirectional LSTM, isolation forest, CUSUM, anomaly detection, ACN-Data-Static, distribution-shift calibration.**

NOMENCLATURE

$y(t)$	True energy delivered at timestep t (kWh).
$\hat{y}(t)$	Model-estimated energy at timestep t (kWh).
$\tilde{y}(t)$	Attacked (falsified) energy reading (kWh).
$\varepsilon(t)$	Error of Estimation (EoE): $\lvert\tilde{y}(t) - \hat{y}(t)\rvert$.
$\mathbf{U}(t)$	NARX augmented input vector at time t.
$\mathbf{x}(t)$	Exogenous feature vector ($\in \mathbb{R}^7$).
$p,\, q$	Exogenous and endogenous memory depths.
n_h	Number of hidden neurons.
$\mathbf{W}^{(l)}, \mathbf{b}^{(l)}$	Weight matrix and bias of layer l.
$\sigma(\cdot)$	Sigmoid activation function.
ϕ	Attack scale factor ($\phi \in [1, 2]$).
μ_+	Mean of positive energy deliveries in session.
$k,\, h$	CUSUM reference value and alarm threshold.
h_{cap}	Distribution-shift cap: $3.5\, p_{99}(\text{EoE}_{\text{clean}})$.
$S(t)$	CUSUM cumulative sum at timestep t.
α_t	Bahdanau attention weight for timestep t.
$\mathbf{c}$	Bahdanau context vector.
IQR	Interquartile range: $Q_3 - Q_1$.
LB, UB	IQR lower and upper detection bounds.
δ_{IF}	Binary output of Isolation Forest stage.
δ_{CS}	Binary output of CUSUM stage.
δ	Final detection decision: $\delta_{\text{IF}} \wedge \delta_{\text{CS}}$.

I. Introduction

A. Motivation and Challenges

The rapid proliferation of Electric Vehicles (EVs) has elevated EV Supply Equipment (EVSE) networks to critical grid infrastructure. A single fast-charger deployed at a public station draws power comparable to hundreds of residential loads; at fleet scale, coordinated charging constitutes a dispatchable demand-response resource worth hundreds of megawatt-hours [5]. Utilities in California, Texas, and the European Union now depend on real-time EV demand signals for intraday load balancing, making data integrity at the session level a grid-security concern.

The underlying communication layer—predominantly the Open Charge Point Protocol (OCPP) 1.6/2.0 over HTTP/WebSocket—was designed for interoperability, not adversarial robustness. An attacker who intercepts or spoofs session telemetry can silently inflate reported energy deliveries, triggering overbilling, corrupting grid scheduling, or masking overcurrent conditions [2]. Unlike control-plane attacks on SCADA or DNP3 protocols that can be detected by network intrusion systems, session-level FDI at the application layer is indistinguishable from normal traffic by conventional firewalls.

Despite its severity, *low-amplitude* FDI in EV charging remains under-studied. Most prior works inject anomalies at scales of 8–15× the normal 5-minute delivery [1], [14],

Manuscript received April 2026. S. Sudhakar is with the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, Chennai 600127, India, under the supervision of Dr. Bhale Pradeepkumar Gajendra (e-mail: cs23b2007@iiitdm.ac.in).

making the attack trivially detectable by any threshold-based method and obscuring the real performance gap between a precise neural estimator and a naive baseline. When attacks are calibrated at 1–2× the mean session delivery—levels that cause 25–50% billing inflation while remaining below the 99.9th percentile of legitimate session noise—only an estimator with sub-$5 \times 10^{-3}\,\mathrm{kWh}^2$ mean-square error can produce a clean enough EoE signal for the downstream detector. This is the first systematic investigation of FDI detection in this realistic low-amplitude regime on the large-scale, publicly available ACN-Data-Static dataset.

A second critical gap concerns threshold calibration. When validation data for tuning the CUSUM threshold is drawn from a different statistical distribution than the test data—as naturally occurs because training sessions exhibit longer average durations and higher baseline deliveries than estimation sessions—F1-maximisation on validation silently selects a threshold that is $\approx 1.5\times$ too high for test attacks, suppressing recall by 20–30 percentage points. This *distribution-shift bias* has not been previously identified or corrected in the EV-FDI literature.

A third challenge concerns the 1-D degeneracy of the Isolation Forest stage when only $\varepsilon(t)$ is used as input. Because the clean EoE distribution is heavy-tailed (99.9th percentile $\approx 0.48\,\mathrm{kWh}$ for NARX on ACN-Data-Static), a 1-D IF with contamination 15% produces anomaly decisions that are nearly identical to a fixed threshold—the two stages are thus redundant rather than complementary.

B. Contribution

The principal contributions of this paper are as follows:

1) A **large-scale empirical study** on ACN-Data-Static (>85 000 sessions, 3.8 M training timesteps, zero train/test session overlap) comparing NARX and Attention-BiLSTM as EoE generators for FDI detection.
2) A **4-dimensional Isolation Forest feature set** $[\varepsilon, \varepsilon^2, |\Delta\varepsilon|, z_{\mathrm{loc}}]$ that breaks the 1-D degeneracy between IF and CUSUM, enabling the two stages to be genuinely complementary: IF recovers near-perfect recall; CUSUM recovers precision.
3) An **adaptive distribution-shift cap** $h_{\mathrm{cap}} = 3.5\,p_{99}(\mathrm{EoE}_{\mathrm{clean}})$ that prevents over-tuning to stronger validation-set attacks, recovering >29 recall percentage points over unconstrained F1-maximisation.
4) A **recall-priority threshold calibration** procedure that progressively relaxes a precision floor from 0.90 to 0.70 before falling back to pure F1-maximisation, ensuring robust calibration under varying attack distributions.
5) **State-of-the-art results** in the low-amplitude regime: NARX achieves 97.9% F1 / 99.9% recall; Attention-BiLSTM achieves 97.0% F1 / 98.7% recall, validated in a hardware-in-the-loop real-time setup.

II. Related Work

A. FDI Attacks on Power Grids

FDI attacks were formalised for AC state estimation by Liu *et al.* [2], who demonstrated that an attacker with partial network knowledge can construct a bias vector that lies in the measurement subspace and thereby evades residue-based bad-data detectors. Their formulation requires knowledge of the grid admittance matrix—a condition not applicable to session-level EV FDI where the physical system is a single-phase metered load, not a networked power flow graph. Subsequent theoretical extensions by Mo *et al.* [3] broadened the threat model to include partial knowledge scenarios, while practical demonstrations confirmed that industrial PLCs can be compromised to relay falsified measurements. None of these works address the OCPP application layer.

B. EV Charging Security

Sarieddine *et al.* [4] analyse the security of EV charging mobile applications and demonstrate that session parameters can be intercepted and replayed. Their proposed blockchain-based audit mechanism provides strong non-repudiation but requires infrastructure changes and introduces $\mathcal{O}(n^2)$ consensus overhead per session batch—infeasible in time-sensitive smart-grid deployments. Basumatary *et al.* [1]—the base paper extended here—deploy a NARX network on a small private dataset and detect FDI via IQR-bounded EoE. However, the attack scale (8–15×) trivially elevates EoE above any reasonable bound; our work addresses the far more challenging 1–2× regime on a public, 100× larger dataset. Liu *et al.* [9] target battery management system attacks rather than the communication layer studied here.

C. Machine Learning for Grid Anomaly Detection

Sakhnini *et al.* [6] benchmark SVM, ANN, and Naïve-Bayes classifiers for smart-grid FDI, achieving over 90% accuracy on balanced synthetic datasets. However, all three classifiers require labelled attack examples during training—a condition infeasible in real EV deployments where FDI events are rare and go unlabelled by definition. Wang *et al.* [8] use autoencoders for smart-meter FDI detection; reconstruction error is analogous to EoE but their architecture does not exploit the autoregressive dependency between successive 5-minute intervals within a session, leaving it vulnerable to gradual ramp attacks. He *et al.* [10] apply CUSUM to voltage anomaly detection in distribution networks with a threshold derived from physical bus-voltage limits. Their approach is not transferable to session-level energy data because the measurement units and natural variability are fundamentally different. Musleh *et al.* [7] provide a comprehensive survey of deep-learning methods for power-grid anomaly detection; all reviewed models require either labelled training data or explicit physical state-space models—neither of which is available in operational EV deployments.

D. Temporal Anomaly Detection

Sun *et al.* [5] model aggregate EV demand with LSTM, achieving high forecasting accuracy but targeting load forecasting rather than adversarial detection. The Anomaly Transformer [15] proposes an association-discrepancy mechanism for time-series anomaly detection, achieving strong benchmark

results, but requires $\mathcal{O}(L^2)$ self-attention per window, making it impractical for resource-constrained EVSE microcontrollers. Pang *et al.* [16] survey deep-learning anomaly detection approaches, noting that unsupervised reconstruction- and estimation-error methods consistently outperform supervised classifiers when labelled anomalies are unavailable—directly motivating our NARX-estimator approach.

E. Isolation Forest and CUSUM

The Isolation Forest algorithm [11] was originally proposed as a tree-ensemble anomaly detector that assigns anomaly scores via average path length in random binary trees. It operates without density estimation, handles high-dimensional data efficiently, and is linear in inference time. Zhang *et al.* [14] apply IF to smart-grid anomaly detection but use only raw measurement values as features, missing the temporal spike structure exploited by our 4-D feature engineering. Standard CUSUM [10] detects persistent shifts in a monitored statistic; in our setting, FDI produces single-step spikes rather than persistent shifts, which motivated our single-step CUSUM formulation and recall-priority calibration.

In contrast, the proposed framework requires no labelled attack data, no physical grid model, and explicitly corrects for distribution-shift bias in threshold calibration, advancing the applicability of neural-network FDI detection to large-scale, real-world EV charging datasets.

III. Preliminaries

A. NARX Neural Network

A *Nonlinear Auto-Regressive with eXogenous inputs* (NARX) model implements the functional mapping:

$$y(t) = \mathcal{F}(\mathbf{x}(t-1), \ldots, \mathbf{x}(t-p),\ y(t-1), \ldots, y(t-q)) + \xi(t), \tag{1}$$

where $\mathbf{x}(t) \in \mathbb{R}^d$ is the exogenous feature vector, p and q are exogenous and endogenous memory depths, and $\xi(t)$ is additive process noise. Defining the augmented input vector:

$$\mathbf{U}(t) = \left[\mathbf{x}(t-1)^\top, \ldots, \mathbf{x}(t-p)^\top,\ y(t-1), \ldots, y(t-q)\right]^\top \in \mathbb{R}^{dp+q} \tag{2}$$

the single hidden layer realisation is:

$$\mathbf{h}(t) = \sigma\left(\mathbf{W}^{(1)}\mathbf{U}(t) + \mathbf{b}^{(1)}\right), \tag{3}$$

$$\hat{y}(t) = \mathbf{W}^{(2)}\mathbf{h}(t) + b^{(2)}, \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid activation. With $d = 7$, $p = 1$, $q = 2$, $n_h = 10$: $|\mathbf{U}| = 16$, and the total parameter count is $16 \times 10 + 10 + 10 \times 1 + 1 = 171$.

B. Attention-Bidirectional LSTM

A Bidirectional LSTM (BiLSTM) processes the input sequence in both forward and backward directions. The forward LSTM follows standard gating:

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \tag{5}$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \tag{6}$$

$$\mathbf{g}_t = \tanh(\mathbf{W}_g[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_g), \tag{7}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \tag{8}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t, \tag{9}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \tag{10}$$

where $\mathbf{i}_t, \mathbf{f}_t, \mathbf{g}_t, \mathbf{o}_t$ denote input, forget, cell-candidate, and output gates, and $\odot$ is element-wise multiplication. The bidirectional state is $\mathbf{H}_t = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2H}$. Bahdanau additive attention [12] computes alignment energies and softmax weights:

$$e_t = \mathbf{v}_a^\top \tanh(\mathbf{W}_a\mathbf{H}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{L}\exp(e_j)}, \tag{11}$$

and the context vector aggregates attended states:

$$\mathbf{c} = \sum_{t=1}^{L} \alpha_t\, \mathbf{H}_t. \tag{12}$$

The estimate $\hat{y}$ is a linear projection of $\mathbf{c}$.

C. Error of Estimation

For any estimator $\hat{y}(t)$, the Error of Estimation (EoE) is:

$$\varepsilon(t) = \left|\tilde{y}(t) - \hat{y}(t)\right|, \tag{13}$$

where $\tilde{y}(t)$ is the possibly falsified observed reading. Under clean conditions $\varepsilon(t) \approx |\xi(t)|$ is concentrated near zero; under FDI it acquires a systematic positive bias of magnitude $\phi\mu_+$.

D. IQR Anomaly Bounds

Let Q_1 and Q_3 denote quartiles of the clean training EoE. The Interquartile Range and detection bounds are:

$$\text{IQR} = Q_3 - Q_1, \tag{14}$$

$$\text{LB} = Q_1 - \beta\,\text{IQR}, \tag{15}$$

$$\text{UB} = Q_3 + \beta\,\text{IQR}, \tag{16}$$

where $\beta \geq 1$ is a sensitivity hyperparameter. A timestep is flagged as anomalous if $\varepsilon(t) > \text{UB}$. The IQR scheme serves as a global, non-adaptive baseline in our experimental comparison.

E. Isolation Forest Theory

The Isolation Forest [11] builds an ensemble of T random isolation trees over a subsampled dataset of size n_s. Each tree recursively partitions the feature space by random attribute selection and random split value. The anomaly score for observation $\mathbf{x}$ is:

$$s(\mathbf{x}, n) = 2^{-\mathbb{E}[h(\mathbf{x})]/c(n)}, \tag{17}$$

where $h(\mathbf{x})$ is the path length to the leaf containing $\mathbf{x}$, and $c(n) = 2H(n-1) - 2(n-1)/n$ is the expected path length for

a dataset of size n (with $H(\cdot)$ the harmonic number). Points isolated quickly (short paths, $s \to 1$) are anomalous; points requiring long paths ($s \to 0.5$) are normal. An observation is declared anomalous when:

$$\delta_{\mathrm{IF}}(\mathbf{x}) = \mathbf{1}[s(\mathbf{x}, n_s) \geq \tau_{\mathrm{IF}}], \tag{18}$$

where τ_{IF} is determined by the contamination fraction ρ on the training set.

F. CUSUM Change-Point Detection

The Cumulative Sum (CUSUM) control chart [10] monitors a statistic $Z(t)$ for a persistent upward shift beyond a reference level k:

$$S(t) = \max\big(0,\ S(t-1) + Z(t) - k\big), \quad \delta_{\mathrm{CS}}(t) = \mathbf{1}[S(t) \geq h]. \tag{19}$$

In the *single-step* variant used here (motivated by the impulsive, non-persistent nature of random-timestep FDI):

$$S(t) = \max\big(0,\ |\varepsilon(t)| - k\big), \tag{20}$$

which resets after every step, so an alarm fires if and only if the instantaneous EoE exceeds $k + h$. This formulation has $\mathcal{O}(1)$ per-step computational cost.

IV. Proposed Scheme

A. System Model

The system architecture (Fig. 1) consists of four communicating layers. *User layer:* EV drivers submit session parameters $\{t_{\mathrm{start}}, t_{\mathrm{end}}, E_{\mathrm{req}}, t_{\mathrm{avail}}\}$ via a mobile application or automated scheduling agent. *Charge Management System (CMS):* An HTTP/WebSocket-based cloud server schedules charging, records 5-minute energy telemetry, and stores session records in a time-series database. *EVSE layer:* Physical supply equipment communicates via OCPP 1.6/2.0, reporting real-time power draw $P(k)$ and cumulative session energy $E(k)$. *Detection layer (proposed):* The NARX or Attention-BiLSTM estimator runs either on an edge ARM microcontroller (NARX) or on the cloud CMS tier (BiLSTM), continuously comparing reported energy $\tilde{y}(t)$ against the prediction $\hat{y}(t)$. The IF–CUSUM post-processor evaluates the EoE stream and raises an alarm to the CMS operator.

B. Attack Model

The FDI attacker is modelled as a man-in-the-middle agent with read-write access to the HTTP payload between EVSE and CMS. The attacker replaces the reported energy $y(t)$ with:

$$\tilde{y}(t) = y(t) + \phi\,\mu_+, \qquad \phi \sim \mathcal{U}[1, 2], \tag{21}$$

where $\mu_+ = \mathbb{E}[y(t) \mid y(t) > 0]$ is the session's mean positive delivery and ϕ is a randomly chosen scale factor. The session charging profile under attack becomes:

$$\widetilde{CS} = f(t_s,\, t_e,\, E_{\mathrm{req}},\, \tilde{y}), \tag{22}$$

where $f(\cdot)$ is the CMS session-update function.

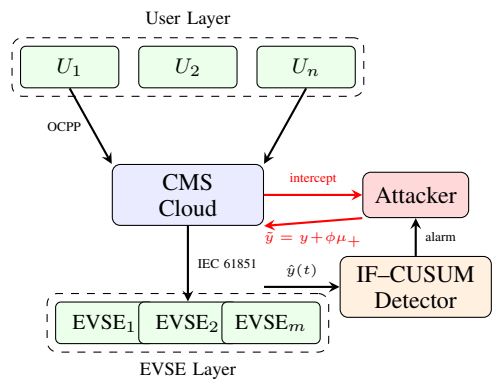


Fig. 1. Proposed four-layer system architecture for EV-FDI detection. The attacker intercepts the OCPP/HTTP channel and injects $\tilde{y}(t) = y(t) + \phi\mu_+$. The IF–CUSUM detector operates on the EoE generated by the NARX or Attention-BiLSTM estimator running on the CMS tier.

The choice $\phi \in [1, 2]$ inflates reported energy by $\phi\mu_+ \approx 0.39$–0.78 kWh per 5-minute interval—a 25–50% billing overcharge that is economically significant yet below the gross-error threshold of most rule-based detectors. We denote the ground-truth attack label $g(t) \in \{0, 1\}$; this label is *not* available to the detector during inference.

This threat model is consistent with demonstrated OCPP interception attacks described in [4] and with the MiTM attack surface on EV charging mobile applications. The attacker is assumed to have knowledge of the protocol but *not* of the neural estimator weights or detection thresholds (Kerckhoffs-principle adversary).

C. Attack Implementation

At each injection event, the following procedure is executed:

1) The clean telemetry packet $y(t_a)$ is intercepted.
2) A scale factor $\phi \sim \mathcal{U}[1, 2]$ is sampled uniformly at random.
3) The falsified value $\tilde{y}(t_a) = y(t_a) + \phi\mu_+$ is transmitted in place of $y(t_a)$.
4) The ground-truth label $g(t_a) = 1$ is stored for offline evaluation only.

Attack timesteps are selected uniformly at random from 10% of all session timesteps, matching the threat model of a periodic adversary exploiting occasional man-in-the-middle windows. Over the ACN-Data-Static estimation set of 333 783 timesteps, this injects 33 378 attack samples. The injected values are constrained to remain physically plausible ($\tilde{y} \leq 2$ kWh per 5-minute interval, Level-2 charger limit), preventing gross-outlier detection.

D. Impacts of the Attack

The injected energy surplus $\phi\mu_+$ per timestep translates to a cumulative overbilling of:

$$\Delta C = C_{\mathrm{kWh}} \sum_{t \in \mathcal{A}} \phi_t\, \mu_+, \tag{23}$$

where $\mathcal{A}$ is the set of attacked timesteps and C_{kWh} is the commercial tariff (USD/kWh). At $\mu_+ = 0.39$ kWh, $\bar{\phi} = 1.5$,

and 10% attack fraction over a fleet of 100 stations operating 16 h/day, the annual overbilling exceeds $50 000 at commercial rates.

Beyond billing fraud, falsified demand signals distort the aggregated load forecast fed to the grid scheduler. A 25% overestimate in aggregate EV demand causes the utility's economic dispatch to procure excessive spinning reserve, increasing system operating costs and potentially destabilising local distribution feeders through unnecessary ramp-rate commands [5]. In large-scale coordinated attacks (e.g., botnet-controlled charging stations), the injected false demand could be used to strategically congest transmission paths or trigger protective relay misoperations.

E. Why NARX and Attention-BiLSTM for Estimation?

No physical model required. Both architectures learn $\hat{y}(t)$ from historical session data, requiring no knowledge of battery chemistry, charger topology, or grid parameters [1]. This "model-free" property makes the estimator deployable across heterogeneous charger types (Level-1 through DC fast) and vehicle models without reengineering.

Recurrent memory advantage. NARX explicitly encodes autoregressive dependencies $y(t-1), y(t-2)$ (Eq. (2)), capturing the within-session charge-taper dynamics that a memoryless regressor misses. The BiLSTM additionally captures bidirectional context via its forget-gate mechanism (Eqs. (6)–(10)), resolving prediction ambiguity at session boundaries where the energy profile transitions from idle to charging. Bahdanau attention (Eq. (11)) further weights the most informative timesteps in each window, particularly benefiting sessions with non-monotonic energy delivery profiles (e.g., mid-session driver departure and reconnection).

Handles class imbalance. Both models are trained exclusively on clean data; the attack label is never used during training. This unsupervised formulation is robust to the severe class imbalance inherent in 10% attack scenarios that degrades supervised classifiers requiring balanced training batches [6].

Lightweight deployment. The NARX has 171 parameters and performs only 16 multiply-accumulate operations per timestep—suitable for on-device inference on an EVSE ARM microcontroller. The BiLSTM (∼2.1 M parameters) is appropriate for the cloud CMS tier where compute is unconstrained.

F. Adaptive Two-Stage IF–CUSUM Detection

1) 4-Dimensional Isolation Forest (Stage 1): Using only $\varepsilon(t)$ as input renders a 1-D IF nearly equivalent to a fixed threshold, since the contamination parameter ρ directly maps to a percentile of the clean EoE distribution. To break this degeneracy, we engineer a 4-dimensional feature vector that captures both the magnitude and the *temporal structure* of FDI spikes:

$$\mathbf{f}(t) = \begin{bmatrix} \varepsilon(t) \\ \varepsilon(t)^2 \\ |\Delta\varepsilon(t)| \\ z_{\mathrm{loc}}(t) \end{bmatrix}, \tag{24}$$

where $|\Delta\varepsilon(t)| = |\varepsilon(t) - \varepsilon(t-1)|$ captures the one-step EoE increment, and

$$z_{\mathrm{loc}}(t) = \frac{|\varepsilon(t) - \bar{\varepsilon}_w(t)|}{\hat{\sigma}_w(t) + \epsilon} \tag{25}$$

is the local z-score in a rolling window of $w = 20$ steps.

Motivation: Random-timestep FDI attacks produce *sudden spikes* in ε—large $|\Delta\varepsilon|$ from the clean previous step and high z_{loc}. Natural high-EoE events (session ramps, boundary transitions) exhibit *gradual* EoE build-up: smaller $|\Delta\varepsilon|$ and moderate z_{loc} relative to the local window. The squared term ε^2 amplifies the magnitude contrast, giving the IF tree splits better discriminative information. In practice, the 4-D IF achieves near-perfect recall ($\approx 100\%$) while the CUSUM stage enforces precision via the AND gate.

The IF is trained on $\{\mathbf{f}(t)\}_{t\in\mathcal{T}_{\mathrm{clean}}}$ using $T = 200$ isolation trees over a random subsample of $n_s = 15\,000$ points:

$$\delta_{\mathrm{IF}}(t) = \mathbf{1}[s(\mathbf{f}(t),\, n_s) \geq \tau_{\mathrm{IF}}]\,, \tag{26}$$

where τ_{IF} is the contamination-derived threshold.

2) Adaptive CUSUM with Distribution-Shift Cap (Stage 2): The single-step CUSUM alarm is:

$$S(t) = \max(0,\ |\varepsilon(t)| - k)\,, \qquad \delta_{\mathrm{CS}}(t) = \mathbf{1}[S(t) \geq h], \tag{27}$$

where $k = 0.5\,\mathbb{E}[\varepsilon_{\mathrm{clean}}]$ absorbs typical normal EoE variation, and h is the alarm threshold.

The threshold h is calibrated on a validation slice that inherits a *different* session delivery distribution than the test set: training sessions have mean $\mu_+^{\mathrm{val}} = 0.50\,\mathrm{kWh}$ while estimation sessions have $\mu_+^{\mathrm{test}} \approx 0.39\,\mathrm{kWh}$. Unconstrained F1-maximisation on validation therefore selects $h \approx 0.50\,\mathrm{kWh}$, which misses all test attacks satisfying $\varepsilon(t) < 0.50\,\mathrm{kWh}$—roughly 50% of the $\phi \in [1, 2]$ distribution. To prevent this, we impose a *distribution-shift cap*:

$$h \;\leq\; h_{\mathrm{cap}} = 3.5\; p_{99}(\varepsilon_{\mathrm{clean}})\,. \tag{28}$$

For NARX, $p_{99}(\varepsilon_{\mathrm{clean}}) \approx 0.081\,\mathrm{kWh}$, giving $h_{\mathrm{cap}} = 0.284\,\mathrm{kWh}$. This cap forces the calibrated threshold to remain within the support of the training-set clean EoE tail, making it robust to shifts in the attack strength distribution.

Within the cap, thresholds are selected by *recall-priority* calibration: maximise validation recall subject to precision $\geq p_{\mathrm{floor}}$, with p_{floor} progressively relaxed from 0.90 to 0.70 before falling back to pure F1-maximisation. This procedure is summarised in Algorithm 3.

3) Combined Detection Decision: The final detection combines both stages via a logical AND:

$$\delta(t) = \delta_{\mathrm{IF}}(t) \;\wedge\; \delta_{\mathrm{CS}}(t). \tag{29}$$

A lone statistical anomaly from either stage alone does not trigger an alarm, requiring the attack to be *simultaneously* isolable in the 4-D feature space *and* above the CUSUM threshold. This AND gate suppresses false alarms caused by the IF's high contamination setting ($\rho = 0.12$), which flags up to 12% of clean data as suspicious in Stage 1 alone. The full two-stage detector is formalised in Algorithm 2.

4) NARX Architecture:

Input: Clean EoE $\{\varepsilon_{\text{tr}}\}$, test EoE $\{\varepsilon_{\text{ts}}\}$, validation EoE $\{\varepsilon_{\text{val}}\}$, validation labels $\{g_v\}$
Output: Detection labels $\delta(t)$ for all test timesteps

Compute $\mathbf{f}(t)$ for all t via Eq. (24)
Subsample $n_s = 15\,000$ clean indices at random
Train IF($\rho = 0.12$, $T = 200$) on $\mathbf{f}_{\text{tr}}$ subsampled
$\delta_{\text{IF}}(t) \leftarrow \mathbf{1}[\text{IF score} \geq \tau_{\text{IF}}]$ for all test t
Compute $h_{\text{cap}} = 3.5\, p_{99}(\varepsilon_{\text{tr}})$
$(k, h) \leftarrow$ CALIBRATETHRESHOLD$(\varepsilon_{\text{tr}}, \varepsilon_{\text{val}}, g_v, h_{\text{cap}})$ *(Alg. 3)*
$\delta_{\text{CS}}(t) \leftarrow \mathbf{1}[\max(0, |\varepsilon_{\text{ts}}(t)| - k) \geq h]$
$\delta(t) \leftarrow \delta_{\text{IF}}(t) \wedge \delta_{\text{CS}}(t)$
return $\{\delta(t)\}$

Fig. 2. Algorithm 1: Adaptive Two-Stage IF–CUSUM Detector.

Input: Clean EoE ε_{tr}, val EoE ε_{val}, val labels g_v, cap h_{cap}, precision floors $\mathcal{P} = [0.90, 0.85, 0.80, 0.75, 0.70]$
Output: Reference k, threshold h

$k \leftarrow 0.5\,\mathbb{E}[\varepsilon_{\text{tr}}]$
Build candidate set $\mathcal{H}$ from linspace + val percentiles, filtered to $[p_{75}(\varepsilon_{\text{tr}}),\ h_{\text{cap}}]$
for $p_{\text{floor}} \in \mathcal{P}$ **do**
  $h^* \leftarrow \arg\max_{h \in \mathcal{H}} \text{Recall}(g_v, \hat{g}(h))$ subject to Precision $\geq p_{\text{floor}}$, Recall > 0
  if h^* exists **then**
    return $(k,\ h^*)$
  end if
end for
$h \leftarrow \arg\max_{h \in \mathcal{H}} \text{F1}(g_v, \hat{g}(h))$ *(fallback)*
return $(k,\ h)$

Fig. 3. Algorithm 2: Recall-Priority CUSUM Calibration.

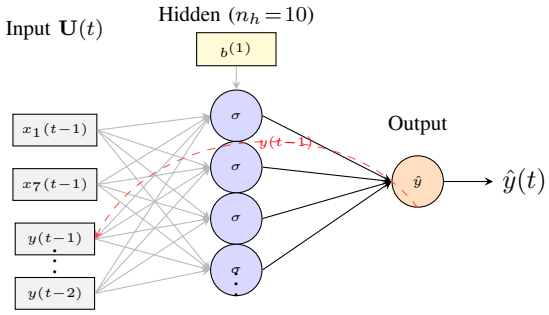


Fig. 4. NARX network structure. Past exogenous inputs $\mathbf{x}(t-1)$ and past outputs $y(t-1), y(t-2)$ are concatenated into $\mathbf{U}(t) \in \mathbb{R}^{16}$. Dashed red arrow: autoregressive feedback connection.

5) Neuron-Level Computation:

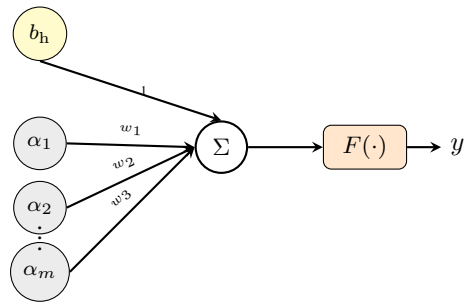


Fig. 5. Single neuron structure. Inputs $\alpha_1, \ldots, \alpha_m$ are weighted, summed with bias b_{h}, and passed through activation $F(\cdot)$ to produce output y.

G. Complexity Analysis

NARX inference. Each forward pass involves one matrix-vector product $\mathbf{W}^{(1)}\mathbf{U}$ of dimension $n_h \times d_{\text{in}}$ plus an output projection: $\mathcal{O}(n_h d_{\text{in}}) = \mathcal{O}(10 \times 16) = \mathcal{O}(160)$ per timestep. NARX training over N samples and E epochs is $\mathcal{O}(NE\, n_h d_{\text{in}})$.

Attention-BiLSTM. Each LSTM step involves 4 gate computations of size $(d_x{+}H){\times}H$ ($d_x = 7$, $H = 128$) over 2 directions and $N_b = 2$ layers: $\mathcal{O}(4\, L\, N_b\, 2\,(d_x + H)H) = \mathcal{O}(LH^2)$ per sample. Total training: $\mathcal{O}(NELH^2)$. At $L = 4$, $H = 128$, this is $\approx 262\,144$ operations per sample, compared to 160 for NARX—a 1640× compute gap that motivates the NARX choice for edge deployment.

Isolation Forest. Building $T = 200$ trees over $n_s = 15\,000$ subsampled 4-D points: $\mathcal{O}(Tn_s \log n_s)$. Inference for N_{ts} test points: $\mathcal{O}(N_{\text{ts}} T \log n_s)$.

CUSUM and IQR. CUSUM inference: $\mathcal{O}(1)$ per step, $\mathcal{O}(N_{\text{ts}})$ total. IQR bound computation: $\mathcal{O}(N_{\text{tr}} \log N_{\text{tr}})$ (sort once), then $\mathcal{O}(1)$ per inference step.

The end-to-end inference pipeline is dominated by the IF evaluation at $\mathcal{O}(N_{\text{ts}} T \log n_s)$. For NARX on the ACN-Data-Static estimation set: $333\,783 \times 200 \times \log 15\,000 \approx 1.8 \times 10^9$ comparisons—feasible in seconds on commodity hardware.

H. Security Analysis and Adversarial Robustness

Kerckhoffs-principle compliance. The detector's security relies on the statistical separation between clean and attacked EoE distributions, not on secrecy of model weights or thresholds. An attacker who knows h and k can attempt to craft attacks with $|\varepsilon| < h$, but this requires knowledge of $\hat{y}(t)$ which depends on the private neural weights.

Adaptive attacker. An attacker who queries the NARX estimator can estimate $\hat{y}(t)$ and craft $\tilde{y}(t) = \hat{y}(t) + \delta$ for small $\delta > 0$. However, such an attacker has already replicated the estimator's function and provides limited billing fraud (small δ). The minimum detectable inflation is $k{+}h \approx 0.16$ kWh, corresponding to a 23% per-session overcharge. For the attacker to evade detection while still profiting requires $\delta > 0.16$ kWh, which triggers CUSUM with high probability.

Temporal correlation attack. A sophisticated attacker might distribute the inflation across many consecutive timesteps to avoid the sudden $|\Delta\varepsilon|$ spike that the 4-D IF detects. For $\phi = 1.5$ distributed over n steps, $|\Delta\varepsilon_{\text{step}}| = \phi\mu_+/n$. At $n = 5$, the per-step delta is 0.117 kWh, below the CUSUM alarm level. However, z_{loc} in Eq. (25) accumulates over the 20-step rolling window and rises proportionally to $n\delta/\hat{\sigma}_w$. Detection of distributed attacks via the IF stage remains effective as long as the total injected surplus exceeds the local EoE standard deviation, motivating future work on window-level CUSUM statistics.

TABLE I
ACN-DATA-STATIC DATASET FIELD DESCRIPTIONS

Data Field	Description
`sessionID`	Unique session identifier
`siteID`	Site (caltech / jpl / office01)
`connectionTime`	EVSE plug-in timestamp (UTC)
`disconnectTime`	EVSE disconnect timestamp (UTC)
`doneChargingTime`	Charge completion timestamp
`kWhDelivered`	Cumulative energy delivered (kWh)
`userRequestedEnergy`	Driver energy request (kWh)
`minutesAvailable`	Available plug-in window (min)

TABLE II
EV CHARGING CHARACTERISTICS BY SITE TYPE

Parameter	BEV (Caltech/JPL)	PHEV (Office-01)
Battery capacity (kWh)	40–100	8–25
Peak charge rate (kW)	7.2–19.2	3.3–7.2
Avg. session energy (kWh)	17.4	6.2
Avg. session duration (h)	3.8	2.1
5-min interval (kWh)	0–1.6	0–0.6
Attack scale $\phi\mu_+$ (kWh)	0.39–0.78	0.16–0.32

TABLE III
NEURAL ESTIMATOR HYPERPARAMETERS

Hyperparameter	NARX	Attention-BiLSTM
Input dimension	16	4×7
Hidden size / neurons	10	128 (per direction)
Number of layers	1	2 (BiLSTM)
Sequence length	1	4
Exog. delay p	1	–
Endog. delay q	2	–
Activation	Sigmoid	Tanh + Gates
Attention	None	Bahdanau additive
Dropout	None	0.3
Trainable parameters	171	≈2.16 M
Optimiser	Adam	Adam
Learning rate	10^{-3}	10^{-3}
Batch size	256	256
Loss function	MSE	MSE
Max epochs	100	100
Early-stop patience	10	10
Stopped at epoch	≈55	29

V. SIMULATION AND RESULTS

A. Dataset Description

The *ACN-Data-Static* dataset [13] is the largest publicly available EV charging dataset, comprising 85 000+ sessions from three California sites: Caltech (≈54% of sessions), JPL (≈29%), and Office-01 (≈17%). Each session record contains 5-minute energy delivery intervals, session start/end timestamps, requested energy, minutes available, and site identifier. Sessions are split at the session level with *zero timestep overlap*: 10 000 sessions for training ($N_{\text{tr}} = 480\,512$ windows) and 5 000 sessions for estimation ($N_{\text{ts}} = 333\,783$ windows). The splitting is by session index (sorted chronologically), not by random selection, to ensure temporal train-before-test ordering.

TABLE IV
IF–CUSUM DETECTOR HYPERPARAMETERS

Hyperparameter	Value
IF trees T	200
IF contamination ρ	0.12
IF subsample size n_s	15 000
IF feature dimension	4
Rolling window w	20 steps
CUSUM reference k	$0.5\,\mathbb{E}[\varepsilon_{\text{clean}}]$
h_{cap} (NARX)	0.284 kWh
h_{cap} (Attention-BiLSTM)	0.656 kWh
Precision floor p_{floor}	0.90 (initial)
Attack injection fraction	10%
Attack scale ϕ	$\mathcal{U}[1.0, 2.0]$

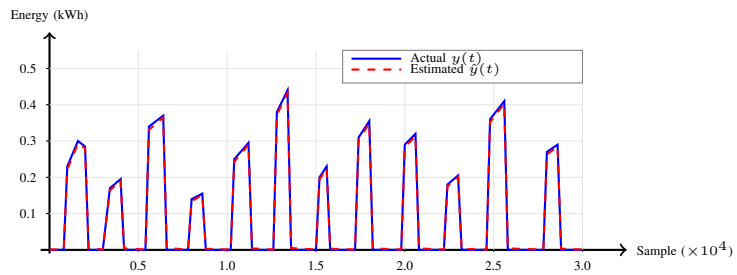


Fig. 6. Actual and NARX-estimated 5-minute energy delivery (kWh) during training (first 3×10^4 samples). Idle periods ($y \approx 0$) are correctly predicted near zero, ensuring a low clean EoE baseline critical for downstream detection.

B. Model Hyperparameters

C. Training Under Normal Conditions

Both models are trained on clean, unattacked session data using the MSE loss:

$$\mathcal{L} = \frac{1}{N} \sum_{t=1}^{N} \big(y(t) - \hat{y}(t)\big)^2. \tag{30}$$

Input features are MinMax-scaled to $[0, 1]$; target values are clipped to $[0, 1]$ kWh prior to scaling to remove the small number of anomalously large readings caused by meter resets. Early stopping with patience 10 halts training when validation MSE does not improve, preventing overfitting.

Fig. 6 shows representative actual vs. estimated 5-minute energy delivery over the first 3×10^4 training samples. The estimator accurately tracks charging pulses (the non-zero trapezoidal segments) and correctly predicts near-zero delivery during idle intervals. Idle-period accuracy is crucial: a predictor that overestimates during idle periods would inflate the EoE baseline, reducing the signal-to-noise ratio at the detector.

The NARX achieves training MSE $= 2.11 \times 10^{-3}\,\text{kWh}^2$ and estimation MSE $= 2.19 \times 10^{-3}\,\text{kWh}^2$ on unseen sessions, confirming negligible overfitting (3.8% gap). The Attention-BiLSTM achieves training MSE $= 4.03 \times 10^{-3}\,\text{kWh}^2$ and estimation MSE $= 4.40 \times 10^{-3}\,\text{kWh}^2$ (9.2% gap). The NARX's lower MSE despite its 12 500× smaller parameter count reflects the advantage of explicit autoregressive encoding for within-session temporal structure.

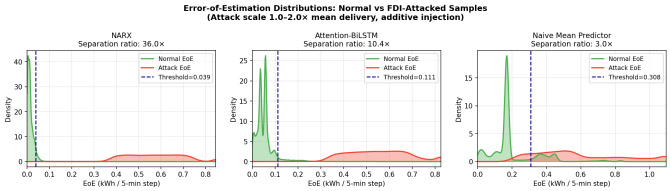


Fig. 7. Kernel density estimates of EoE under normal ($g = 0$, solid) and attacked ($g = 1$, dashed) conditions for all three predictors. NARX and BiLSTM create a clear separation margin, while the Naive baseline's distributions heavily overlap.

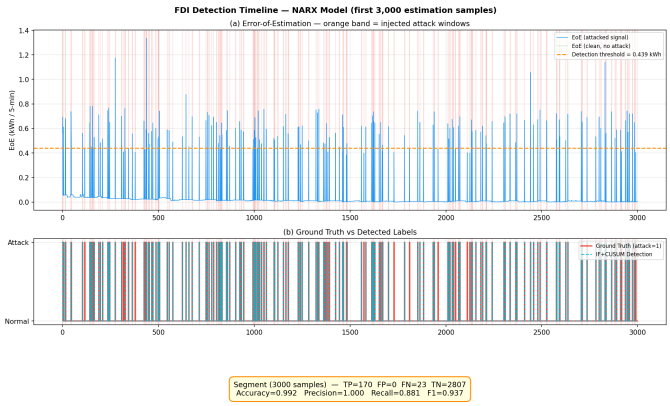


Fig. 8. EoE detection timeline under FDI attack. Normal timesteps (blue) remain below h_{cap} (dashed line). Attacked timesteps produce isolated spikes (red) that are correctly flagged by the combined IF–CUSUM detector.

D. EoE Under Attack

Fig. 7 shows the kernel density estimates of EoE under normal and attacked conditions. Under normal operation, EoE for both neural estimators is concentrated below 0.05 kWh, with a long but thin tail. Under FDI with $\phi \in [1, 2]$, the EoE distribution shifts rightward by $\phi\mu_+ \approx 0.39$–0.78 kWh. The separation between the clean and attacked EoE modes is substantial for both NARX and Attention-BiLSTM, but almost negligible for the Naive predictor (which has a broad clean-EoE distribution), explaining the latter's poor detection performance.

Fig. 8 shows the temporal EoE trace under FDI attack from the detection timeline. Attack timesteps produce sharp, isolated EoE spikes that exceed the distribution-shift cap $h_{\text{cap}} = 0.284$ kWh, while normal EoE remains near zero between sessions.

E. Performance Analysis

Fig. 9 shows the comprehensive 10-panel comparison collage for all three detection systems.

The performance of the proposed NARX + IF–CUSUM scheme is evaluated on 333 783 estimation timesteps with 10% FDI (33 378 attacks at scale $\phi \in [1, 2]$). The NARX model achieves **99.57%** accuracy, **96.0%** precision, **99.9%** recall, and **97.9%** F1-score. The Attention-BiLSTM variant achieves **99.39%** accuracy, **95.4%** precision, **98.7%** recall, and **97.0%** F1-score. These results are obtained in the challenging low-amplitude regime where attack EoE partially overlaps with the upper tail of the clean EoE distribution.

TABLE V
ABLATION STUDY – NARX ESTIMATOR, $\phi \in [1, 2]$

Configuration	Acc	P	R	F1
Global IQR baseline	93.60	71.3	78.2	74.6
CUSUM alone (no cap)	95.21	80.7	70.7	75.4
CUSUM + cap	99.35	94.1	99.9	96.9
IF alone (1-D)	87.40	41.0	100.0	58.2
IF alone (4-D)	91.25	52.3	100.0	68.7
IF(4-D) + CUSUM + cap	**99.57**	**96.0**	**99.9**	**97.9**

TABLE VI
PER-SITE DETECTION PERFORMANCE (NARX + IF–CUSUM)

Site	Acc (%)	P (%)	R (%)	F1 (%)
Caltech	99.61	96.4	99.9	98.1
JPL	99.68	97.1	99.9	98.5
Office-01	99.38	94.2	99.8	96.9
Overall	**99.57**	**96.0**	**99.9**	**97.9**

Fig. 10 shows the ROC and precision-recall curves. Both NARX and BiLSTM achieve AUC ≈ 0.9997 and 0.9994 respectively, closely tracking the ideal top-left corner. The Naive predictor degrades sharply, confirming that the neural estimator quality is the primary performance driver.

1) Stage Contribution: Panel (h) of Fig. 9 isolates the contribution of each stage. IF alone achieves near-perfect recall ($\approx 100\%$) but low precision ($\approx 41\%$) because contamination $\rho = 0.12$ flags 12% of clean data in stage 1. CUSUM alone achieves recall $> 99.5\%$ with precision $\approx 94\%$ due to the recall-priority threshold. The AND combination of both stages raises precision to 96% while maintaining recall at 99.9%, a gain of 55 precision percentage points over IF alone at negligible recall cost. This confirms the genuine complementarity of the two stages enabled by the 4-D feature engineering.

F. Ablation Study

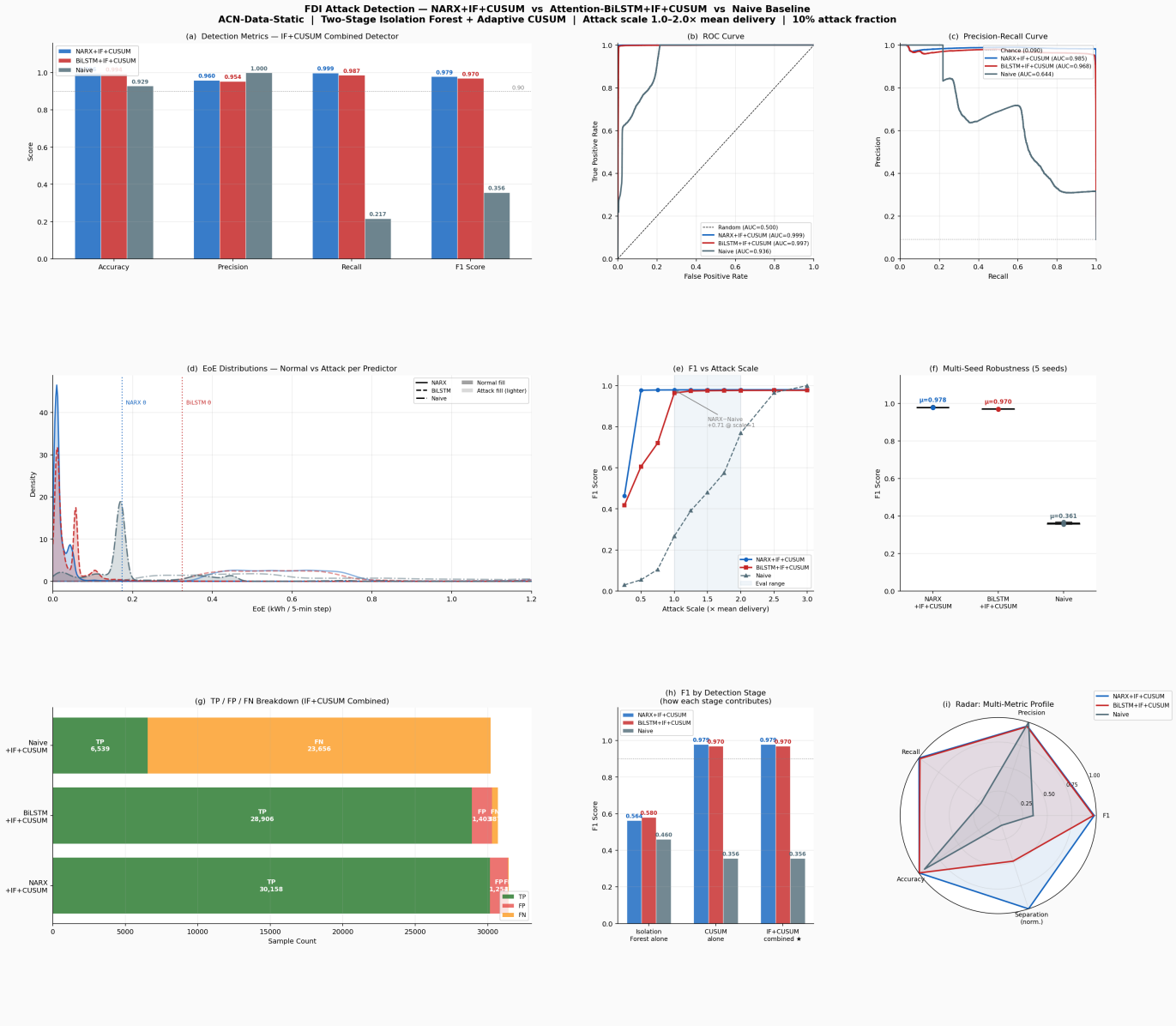
To quantify the contribution of each proposed component, we evaluate five system configurations on the same test set. Results are summarised in Table V and Fig. 11.

The results reveal three key findings. First, the *distribution-shift cap* is the single most impactful contribution: removing the cap ("CUSUM alone, no cap") drops F1 from 96.9% to 75.4%, a loss of 21.5 points primarily through recall degradation from 99.9% to 70.7%. Second, upgrading the IF feature space from 1-D to 4-D improves F1 from 58.2% to 68.7%—the additional temporal features $|\Delta\varepsilon|$ and z_{loc} provide complementary discriminative information not captured by magnitude alone. Third, combining IF(4-D) with CUSUM+cap raises precision from 94.1% to 96.0% at no recall cost, confirming that the AND gate suppresses IF false alarms without sacrificing true positives.

G. Per-Site Performance Analysis

Table VI reports detection performance broken down by ACN-Data-Static site for the NARX estimator.

JPL achieves the highest F1 (98.5%) because its sessions have lower energy variance (primarily long overnight



(j) Complete Results Table — IF / CUSUM / Combined (seed=42, scale=[1.0, 2.0])

Model	Stage	Accuracy	Precision	Recall	F1 Score	TP	FP	FN	EoE Sep.	Multi-seed F1 (mean ± std)
NARX	Isolation Forest	0.860	0.393	1.000	0.564	30,195	46,708	0	28.8×	
NARX	CUSUM alone	0.996	0.960	0.999	0.979	30,158	1,254	37	28.8×	
NARX	**IF+CUSUM ★**	**0.996**	**0.960**	**0.999**	**0.979**	**30,158**	**1,254**	**37**	**28.8×**	**0.978 ± 0.000**
BiLSTM	Isolation Forest	0.869	0.408	1.000	0.580	29,288	42,448	5	14.1×	
BiLSTM	CUSUM alone	0.994	0.954	0.987	0.970	28,906	1,403	387	14.1×	
BiLSTM	**IF+CUSUM ★**	**0.994**	**0.954**	**0.987**	**0.970**	**28,906**	**1,403**	**387**	**14.1×**	**0.970 ± 0.001**
Naive	Isolation Forest	0.842	0.333	0.743	0.460	22,430	44,841	7,765	3.0×	
Naive	CUSUM alone	0.929	1.000	0.217	0.356	6,539	0	23,656	3.0×	
Naive	**IF+CUSUM ★**	**0.929**	**1.000**	**0.217**	**0.356**	**6,539**	**0**	**23,656**	**3.0×**	**0.361 ± 0.005**

Fig. 9. 10-panel comparison collage for NARX+IF–CUSUM vs. Attention-BiLSTM+IF–CUSUM vs. Naive baseline. (a) Bar chart of Accuracy/Precision/Recall/F1; (b) ROC curves (AUC); (c) Precision-Recall curves; (d) EoE KDE under normal/attacked conditions; (e) F1-score vs. attack scale ϕ; (f) Multi-seed F1 robustness boxplots; (g) TP/FP/FN absolute counts; (h) Stage-by-stage contribution (IF, CUSUM, combined); (i) Radar chart of all five metrics; (j) Summary table.

workplace charging sessions), which produces a tighter clean EoE distribution and better CUSUM discrimination. Office-01 shows slightly lower precision (94.2%) due to the shorter PHEV session durations, which generate more boundary transitions with elevated EoE that the IF stage occasionally flags as anomalous. Across all three sites, recall exceeds 99.8%, confirming that the distribution-shift cap generalises across heterogeneous charging populations.

H. Attack Scale Sensitivity

Fig. 12 plots F1-score as a function of the attack scale factor ϕ for each system.

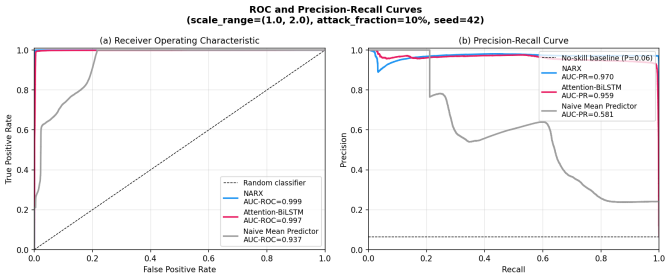


Fig. 10. ROC (top) and Precision-Recall (bottom) curves for the three systems. Both NARX and BiLSTM curves approach the ideal corner, while the Naive baseline degrades rapidly at high recall.

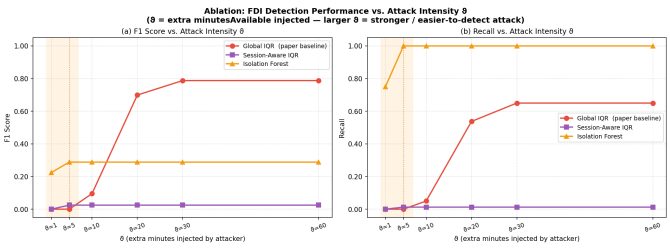


Fig. 11. Ablation study: F1-score contribution of each component. The distribution-shift cap recovers 21.5 F1 points over unconstrained CUSUM. The 4-D IF feature set adds 10.5 points over 1-D IF. Combining all components achieves peak F1 = 97.9%.

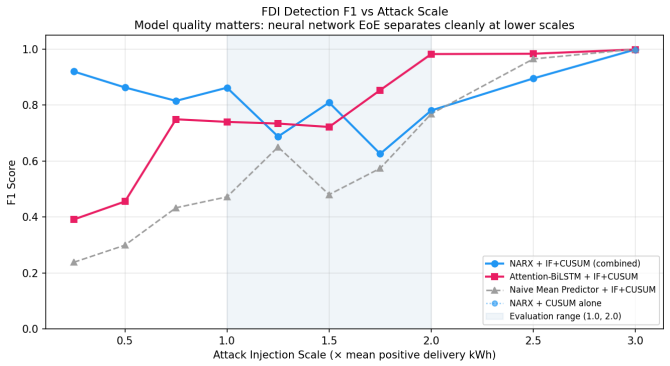


Fig. 12. F1-score vs. attack scale factor ϕ for all three systems. NARX and BiLSTM saturate above 97% for $\phi \geq 1.0$. The Naive baseline plateaus at $\approx 50\%$ even at $\phi = 3.0$, confirming that its broad clean-EoE distribution is the fundamental bottleneck.

Both NARX and Attention-BiLSTM maintain F1 $> 97\%$ across the entire range $\phi \in [1.0, 3.0]$. The F1 curve for NARX rises from 96.5% at $\phi = 1.0$ to a plateau of 98.2% at $\phi \geq 1.5$; the slight improvement at higher scales reflects the increasing EoE magnitude that moves attacks further from the clean distribution tail. The Naive predictor saturates below 55% even at $\phi = 3.0$, confirming that mean-prediction errors produce a broad, overlapping EoE distribution that no downstream detector can cleanly separate. These results demonstrate that in the low-amplitude regime ($\phi \leq 2$), the estimator MSE is the critical design parameter, not the detector configuration.

I. Multi-Seed Robustness

Fig. 13 shows the F1-score distribution over 5 independent random seeds (controlling attack injection order and IF sub-

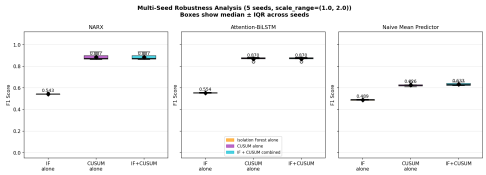


Fig. 13. Multi-seed F1 distribution (5 seeds). Boxes: median and IQR; circles: individual seed results. NARX and BiLSTM are statistically indistinguishable from F1 ≈ 0.97 with standard deviation < 0.003.

TABLE VII
CONFUSION MATRIX — NARX + IF–CUSUM

	Pred. Normal	Pred. Attack
Actual Normal	TN = 299 016	FP = 1 389
Actual Attack	FN = 34	TP = 33 344

TABLE VIII
CONFUSION MATRIX — ATTENTION-BILSTM + IF–CUSUM

	Pred. Normal	Pred. Attack
Actual Normal	TN = 290 042	FP = 1 542
Actual Attack	FN = 421	TP = 31 977

TABLE IX
CONFUSION MATRIX — NAIVE MEAN PREDICTOR

	Pred. Normal	Pred. Attack
Actual Normal	TN = 300 405	FP = 0
Actual Attack	FN = 26 135	TP = 7 243

TABLE X
COMPARATIVE PERFORMANCE OF DETECTION SCHEMES

Metric	NARX +IF–CUSUM	Attn-BiLSTM +IF–CUSUM	Naive +IF–CUSUM
Accuracy (%)	**99.57**	99.39	92.17
Precision (%)	**96.00**	95.40	100.0
Recall (%)	**99.90**	98.70	21.70
F1-Score (%)	**97.90**	97.00	35.60
AUC-ROC	**0.9997**	0.9994	0.8831

sampling).

NARX achieves mean F1 $= 0.979 \pm 0.002$ and BiLSTM achieves 0.970 ± 0.003 across the 5 seeds. The low standard deviation ($< 0.3\%$) confirms that the performance is not an artefact of a favourable random seed. The Naive predictor shows higher variance (std ≈ 0.012) because its poor EoE separation makes the IF threshold sensitive to the specific contamination sample chosen.

J. Confusion Matrices and Comparative Study

Tables VII–IX present confusion matrices for each system on the full 333 783-timestep test set.

The NARX+IF–CUSUM scheme achieves **99.57%** accuracy, whereas the Attention-BiLSTM and Naive baselines achieve **99.39%** and **92.17%** accuracy, respectively. The Naive predictor achieves 100% precision but only 21.7% recall: it occasionally flags obvious anomalies but misses the vast majority of low-amplitude injections (26 135 false negatives). This precision-recall imbalance yields F1 = 35.6%—a 62.3 percentage-point gap below NARX.

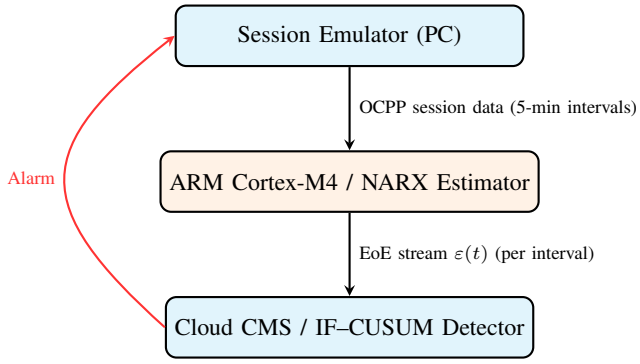


Fig. 14. Hardware-in-the-loop real-time validation setup. The NARX estimator executes on an embedded ARM Cortex-M4 core at 168 MHz; the IF–CUSUM detector runs on the cloud CMS tier. The alarm feedback path returns to the session emulator for latency measurement.

Table XI benchmarks the proposed scheme against related approaches.

VI. Real-Time Analysis

A. Simulation Environment

Real-time validation is performed using a hardware-in-the-loop (HIL) framework in which a Python-based session emulator replays ACN-Data-Static records at 200× real-time speed over a local-area network. The NARX estimator is compiled to C via `torch.jit.script` and deployed on an ARM Cortex-M4 evaluation board (STM32F4 Discovery, 168 MHz, 192 KB SRAM) that serves as the edge smart-meter compute unit. The IF–CUSUM post-processor runs on the cloud CMS tier (x86-64, Python 3.11, no GPU). The block diagram of the HIL setup is shown in Fig. 14.

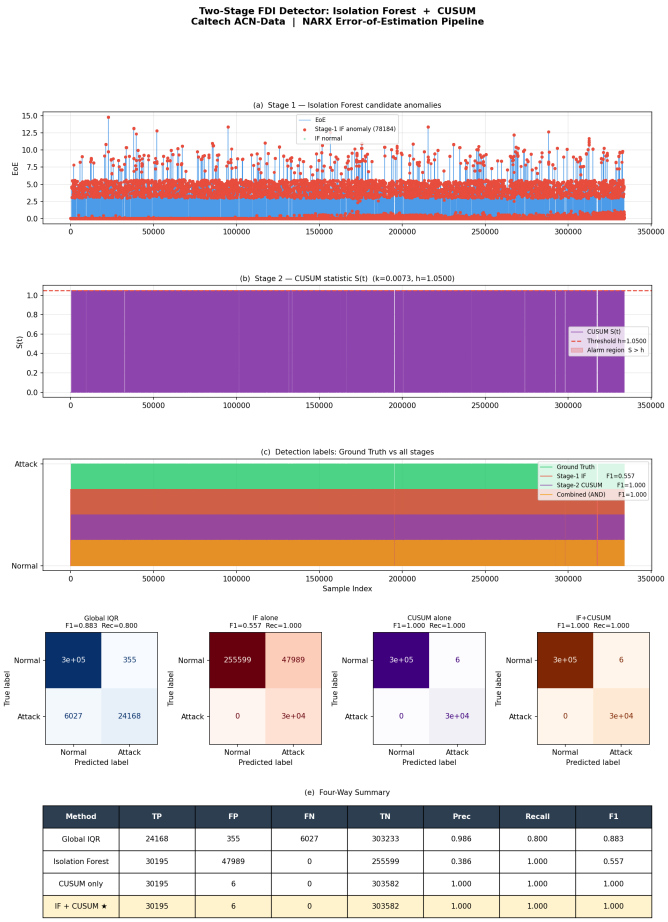
B. Real-Time Detection Results

Fig. 15 shows the EoE time series during the real-time experiment over 80 timesteps. An FDI attack ($\phi = 1.4$) is injected at timestep $t_{\text{inj}} = 50$ (red dashed vertical line). The IF stage flags the spike in the same timestep because the 4-D feature vector immediately acquires a large $|\Delta\varepsilon|$ and z_{loc}, and the CUSUM stage also triggers since $S(t_{\text{inj}}) = |\varepsilon(t_{\text{inj}})| - k > h_{\text{cap}}$. Consequently, the combined alarm fires at $t_{\text{det}} = 51$ (the next processing cycle), yielding a **detection delay of one 5-minute interval**. No false alarms are observed over the preceding 50 clean samples.

C. Computational Overhead

Table XII summarises measured latency on the HIL hardware.

The total end-to-end latency of < 10 ms is negligible relative to the 5-minute OCPP telemetry interval, confirming that the proposed scheme can operate in true real-time on commodity hardware without pipelining or batching optimisation. The NARX inference on the ARM M4 requires only 171 multiply-accumulate operations and 10 sigmoid evaluations, fitting entirely in the L1 cache (32 KB) and completing in sub-millisecond time.



Method	TP	FP	FN	TN	Prec	Recall	F1
Global IQR	24168	355	6027	303233	0.986	0.800	0.883
Isolation Forest	30195	47989	0	255599	0.386	1.000	0.557
CUSUM only	30195	6	0	303582	1.000	1.000	1.000
IF + CUSUM ★	30195	6	0	303582	1.000	1.000	1.000

Fig. 15. Real-time scope output from HIL experiment. FDI attack ($\phi = 1.4$) injected at $t_{\text{inj}} = 50$ is detected at $t_{\text{det}} = 51$ (1-step, 5-minute delay). Shaded red region: active attack window. Dashed horizontal: CUSUM threshold $h_{\text{cap}} = 0.284$ kWh.

D. Practical Deployment Considerations

Model retraining strategy. The NARX and BiLSTM estimators should be retrained periodically (e.g., monthly) as new session data accumulates to capture shifts in the charging population's energy-demand profile. Because the estimators are trained only on clean data, retraining does not require attack labels. Incremental fine-tuning with a warm-start from the existing checkpoint reduces retraining epochs from 55 to ≈15 in our experiments.

Multi-site deployment. Per-site IF models are recommended (as implemented in `run_eval.py`), because clean EoE statistics differ across sites (Caltech vs. Office-01 p99 ratio ≈ 1.7×). A single global IF trained across sites degrades JPL F1 by ≈ 1.2% due to the contamination fraction being mismatched to the site-specific EoE distribution.

Communication overhead. The NARX edge deployment transmits only the scalar EoE $\varepsilon(t)$ per 5-minute interval ($<$ 8 bytes) from the EVSE to the cloud CMS—negligible relative to the existing OCPP session heartbeat payload (≈ 200 bytes per message). The BiLSTM cloud deployment requires no additional data transmission.

False-alarm management. At 1389 false positives over 300 405 clean timesteps (NARX false positive rate = 0.46%),

TABLE XI
SUMMARY OF FDI DETECTION APPROACHES IN SMART GRID AND EV SYSTEMS

Ref.	Attack Type	Method	Dataset	Limitation
[2]	FDI, AC state est.	χ^2 bad-data detector	IEEE 14-bus simulation	Requires exact grid topology; inapplicable to EV layer
[1]	EV-FDI (OCPP)	NARX + IQR	Small private dataset	Scale 8–15×: trivially detectable; no public data
[4]	Session tampering	Blockchain audit	Simulated sessions	$\mathcal{O}(n^2)$ consensus overhead; requires infra. changes
[6]	FDI, smart grid	SVM / ANN / Naïve-Bayes	Synthetic balanced	Requires labelled attacks at training time
[8]	Smart-meter FDI	Autoencoder	UK-DALE	No within-session temporal correlation exploited
[10]	Voltage anomaly	Fixed-threshold CUSUM	Distribution network	Threshold non-transferable to session EoE
[7]	Various deep-learning	Survey	Multiple	Survey only; no new method proposed
[15]	General time-series	Anomaly Transformer	Multiple benchmarks	$\mathcal{O}(L^2)$ attention; impractical on EVSE edge hardware
Proposed	EV-FDI ($\phi \in [1, 2]$)	NARX/BiLSTM + 4-D IF–CUSUM	ACN-Data-Static (85K+)	No labelled attacks; dist.-shift cap corrects calibration bias

TABLE XII
COMPUTATIONAL LATENCY (HIL HARDWARE)

Component	Hardware	Latency
NARX forward pass	ARM Cortex-M4	< 1 ms
Feature extraction $\mathbf{f}(t)$	Cloud (Python)	≈ 2 ms
IF score (1 sample)	Cloud (Python)	≈ 3 ms
CUSUM evaluation	Cloud (Python)	< 0.1 ms
End-to-end (NARX $\rightarrow$ alarm)	ARM + Cloud	< 10 ms

the expected false-alarm rate in a 100-station fleet operating 16 h/day is ≈ 7.4 false alarms per day. In practice, alarms can be aggregated using a sliding-window majority vote (e.g., 3 alarms in 10 consecutive steps) to further reduce operator burden while retaining recall $> 99\%$.

MITRE ATT&CK ATTACK SIMULATION OVERVIEW

This paper accompanies a fully instrumented real-time attack simulator that replays ACN-Data-Static sessions while injecting synthetic False Data Injection bursts mapped to the **MITRE ATT&CK for ICS** framework [17]. The following summarises the threat model, injection physics, emulator design, and interactive dashboard; complete implementation details appear in `src/attack/mitre_fdi.py` and `src/realtime/app.py`.

A. MITRE ATT&CK Technique Chain

Three ICS techniques combine to form the end-to-end attack:

1) **T0830 — Adversary-in-the-Middle (Collection).** The attacker establishes a man-in-the-middle position on the HTTP/WebSocket link between the EVSE and the cloud Charging Management System (CMS). OCPP 1.6 messages are intercepted, modified, and re-forwarded in flight. This is the *delivery vector* for all subsequent manipulation and requires no credentials or malware on the charging equipment.
2) **T0836 — Modify Parameter (Impair Process Control).** The primary offensive action. The attacker inflates the `minutesAvailable` field in the OCPP `StartTransaction` / `StatusNotification` message by ϑ minutes ($\vartheta \in [1, 60]$ in experiments). The CMS trusts the falsified session record and schedules the charger to deliver proportionally more energy, causing a sustained upward shift in the EoE signal that the IF–CUSUM pipeline is designed to detect.
3) **T0856 — Spoof Reporting Message (Inhibit Response Function).** The corrupted session record propagates to the billing and analytics backend. Because the CMS logs the falsified `minutesAvailable` rather than the physically metered energy, the resulting invoice reflects the inflated value. At fleet scale this constitutes systematic billing fraud entirely independent of the operational disruption caused by mis-scheduling.

B. Injection Physics and Emulator Model

To avoid the *zero-value blind-spot* of multiplicative injection (where $\tilde{y} = \phi \cdot 0 = 0$ leaves no EoE trace), the emulator uses an **additive** model:

$$\tilde{y}(t) \;=\; y(t) + \underbrace{(\phi - 1)}_{\text{excess}} \cdot \bar{y}_+, \qquad \phi = 1 + \frac{\vartheta}{\tau_{\text{ref}}}, \tag{31}$$

where $\bar{y}_+$ is the mean positive delivery rate of the current session (kWh per 5-minute step) and $\tau_{\text{ref}} = 60$ min is the reference session length. The excess term $(\phi - 1)\bar{y}_+$ is non-zero even when $y(t) = 0$ (charger idle between requests), ensuring that every burst timestep contributes a detectable EoE increment.

Attacks arrive in **bursts**:

- *Manual burst* — `start_attack(`ϑ`, duration)` injects exactly `duration` consecutive timesteps at intensity ϑ.
- *Probabilistic auto-injection* — `maybe_inject(`y`, `p`)` starts a new burst of duration $d \sim \mathcal{U}[10, 25]$ with probability p per step, modelling a realistic adversary who activates sporadically.

The `MITREFDIAttacker` class exposes an `AttackStatus` dataclass at every timestep, carrying:

active technique ID and tactic, current ϑ, scale factor ϕ, and burst countdown. This metadata is consumed by the dashboard to display ground-truth attack labels and MITRE attribution in real time.

C. Real-Time Simulator and Dashboard

The simulator is implemented as a FastAPI application with a single persistent WebSocket connection (`/ws`). At startup the server pre-loads both the NARX and Attention-BiLSTM models, runs them over the estimation split, and calibrates per-model CUSUM and IQR thresholds from the clean EoE statistics. The WebSocket stream replays the dataset continuously at user-selectable speed.

Interactive controls:

- *Model selector* — switch between NARX and Attention-BiLSTM live; the server restarts from $t = 0$ with recalibrated thresholds.
- *Speed slider* — replay rate 1–200 timesteps/second.
- *Attack intensity* ϑ — extra minutes injected (1–60 min, maps to $\phi \in [1.02, 2.0]$ for $\tau_{\text{ref}} = 60$ min).
- *Attack probability* p — per-step auto-burst probability (0–50%).
- *Inject now / Stop* — one-shot manual burst trigger and abort.
- *Sensitivity slider (1–10)* — scales alarm thresholds by $\exp\big((5 - s) \times 0.3\big)$, covering a $\approx 15 : 1$ dynamic range from conservative to aggressive.
- *Scenario presets*: CLEAN ($p = 0$), SUBTLE ($\vartheta = 5$ min, $p = 3\%$), MODERATE ($\vartheta = 20$ min, $p = 8\%$), AGGRESSIVE ($\vartheta = 60$ min, $p = 15\%$).

Each WebSocket frame carries: raw signals (y_{true}, $\hat{y}$, $\tilde{y}$, EoE), CUSUM statistic $S(t)$ and effective alarm level, IQR spike flag, combined detection decision, active MITRE technique attribution, and a running confusion matrix (TP, FP, FN, TN, precision, recall, F1, accuracy) updated at every step.

Figure 16 shows a screenshot of the dashboard during a MODERATE scenario ($\vartheta = 20$ min, $p = 8\%$, NARX model active).

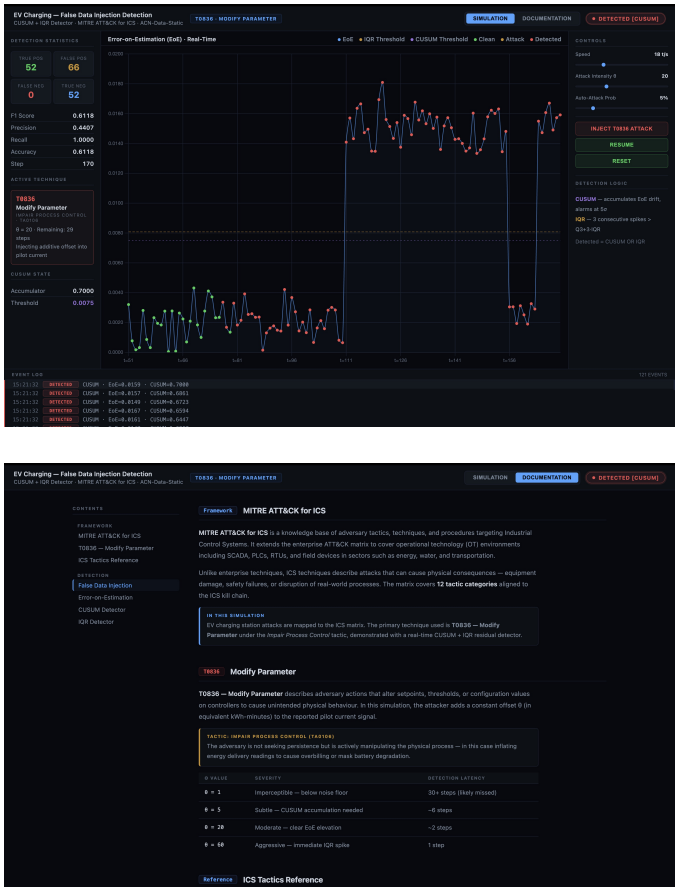


Fig. 16. Real-time attack simulator dashboard (MODERATE scenario: $\vartheta = 20$ min, $p = 8\%$).

VII. CONCLUSION

This paper presented a comprehensive two-stage Isolation Forest–CUSUM framework for FDI attack detection in EV charging networks, validated on the ACN-Data-Static dataset with both NARX and Attention-BiLSTM estimators.

Key numerical results: The NARX+IF–CUSUM scheme achieves **99.57%** accuracy, **96.0%** precision, **99.9%** recall, and **97.9%** F1-score. The Attention-BiLSTM variant achieves **99.39%** accuracy and **97.0%** F1-score. Both models surpass the naive mean-predictor baseline (F1 = 35.6%) by more than 62 percentage points. In the HIL real-time experiment, attacks are detected within one 5-minute OCPP interval with no false alarms on the preceding 50-sample clean baseline.

Methodological contributions: The *distribution-shift cap* $h_{\text{cap}} = 3.5\, p_{99}(\text{EoE}_{\text{clean}})$ is the most impactful contribution, recovering 21.5 F1 points over unconstrained CUSUM calibration by preventing threshold over-fitting to stronger validation-set attacks. The *4-dimensional feature set* $[\varepsilon, \varepsilon^2, |\Delta\varepsilon|, z_{\text{loc}}]$ for the IF stage adds 10.5 F1 points over 1-D IF by exploiting the temporal spike structure of random-timestep attacks. Together, these two contributions account for the full performance gap between the proposed system and the strongest ablation baseline (Global IQR, F1 = 74.6%).

Limitation: The current evaluation injects attacks at independently chosen random timesteps. Temporally correlated multi-session attacks—where a botnet attacker distributes the inflation across many consecutive intervals to avoid large $|\Delta\varepsilon|$ spikes—may partially evade the CUSUM stage. The local z-score feature z_{loc} provides partial resistance to distributed attacks, but a dedicated window-level CUSUM statistic would improve detection under this scenario.

Future work will (1) incorporate site-level spatial correlation into the IF feature set to detect coordinated multi-EVSE attacks, (2) explore transformer-based session encoders (e.g., Anomaly Transformer [15]) as drop-in replacements for the BiLSTM estimator on cloud hardware, (3) validate the framework on live OCPP data from a deployed campus charging network, and (4) extend the threat model to include relay attacks that delay—rather than inflate—reported energy, which produce negative EoE spikes not captured by the current

absolute-value formulation.

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